23450

C23-M-MRAC–406

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BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

DME – FOURTH SEMESTER EXAMINATION

PRODUCTION DRAWING

Time : 3 Hours] [*Total Marks* : 60

PART—A 5×4=20

Instructions : (1) Answer **all** questions.

(2) Each question carries **five** marks.

(3) Draw the following neatly with proportionate dimensions.

(4) Use of Production Drawing limits, fits and tolerance tables are allowed.

1. The dimensions of a shaft and hole are given below :

Hole : 44·500 mm, 44·515 mm   Shaft : 43·975 mm, 43·957 mm

Find the following :

(a) Tolerance of shaft

(b) Tolerance of hole

(c) Max. allowance

(d) Min. allowance

(e) Type of fit

2. Draw the symbols for the following characteristics to be toleranced.

(a) Parallelism

(b) Position

(c) Symmetry

(d) Concentricity

(e) Straightness

3. Give the range of surface roughness values in microns for the following manufacturing process.

(a) Die casting

(b) Hand grinding

(c) Shaping

(d) Boring

(e) Polishing

4. Write the meanings for the following code designations of materials/ standard mechanical components.

(a) Fe470 W

(b) 75T5

(c) Hexagon head Bolt M20 × 70, –4·6

(d) Oil seal B25 × 40 × 7, IS : 5129

(e) Spline shaft 6 × 23 × 6

PART—B

40

Instructions : (1) Answer *any one* of the following questions.

(2) Each question carries **forty** marks.

(3) All dimensions are in mm.

(4) Choose suitable scale.

5. Study the given assembly drawing of Screw Jack. 20+5+3+3+4+5=40

(a) Draw the part drawings.

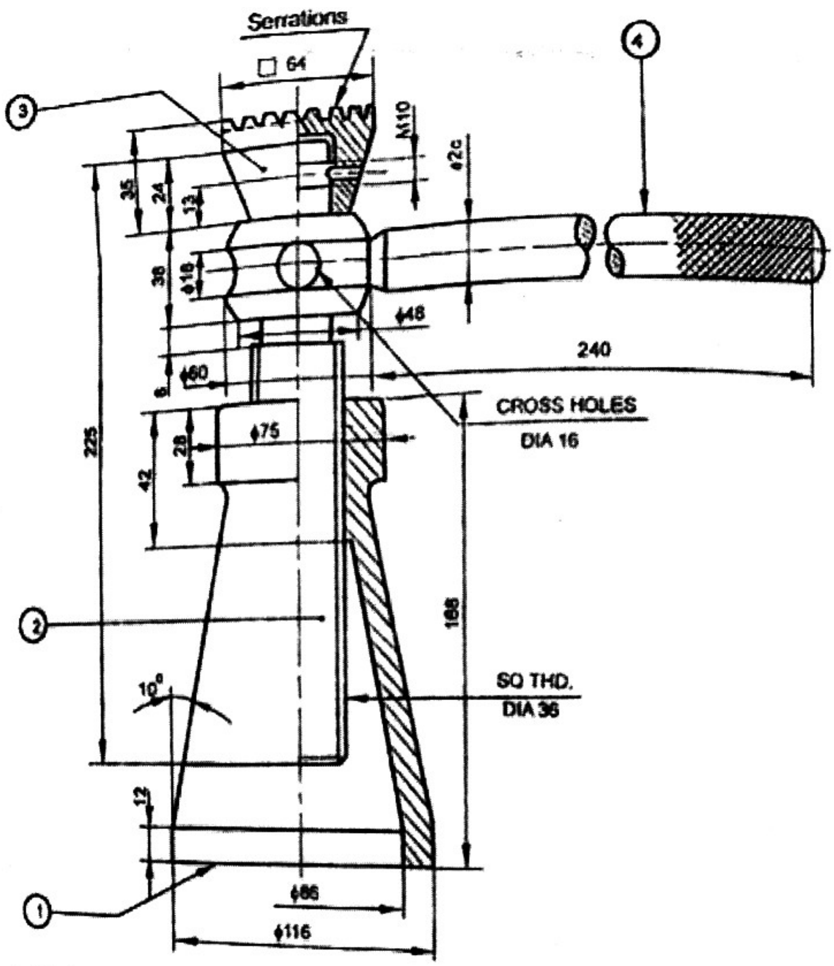
(b) Indicate the suitable fits and dimensional tolerances wherever required.

(c) Apply Geometrical tolerances wherever needed.

(d) Indicate surface roughness values/symbols to the components.

(e) Prepare bill of materials.

(f) Prepare process sheet for the manufacturing of "Screw".



Parts List

Part. No	Name	Raw material	Qty.
1	Body	C.I Casting	1
2	Screw	MCS Φ63 Bar stock	1
3	Cup	C.I -Casting	1
4	Tommy bar	MS Φ20 Bar stock	1

6. Study the given assembly drawing of Foot-Step Bearing. 20+5+3+3+4+5=40

(a) Draw the part drawings.

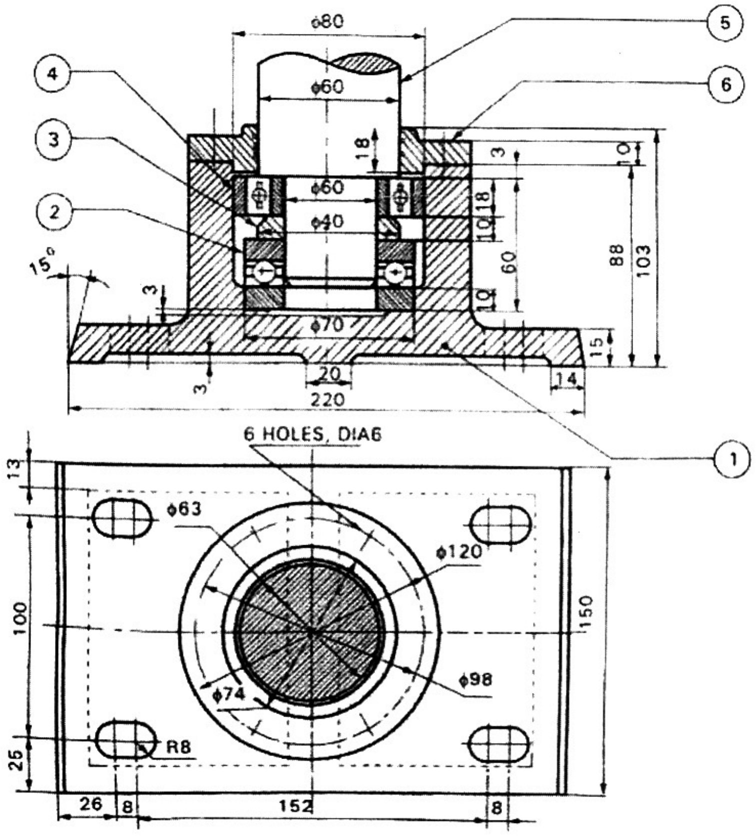
(b) Indicate the suitable fits and dimensional tolerances wherever required.

(c) Apply geometrical tolerances wherever needed.

(d) Indicate surface roughness values/symbols to the components.

(e) Prepare bill of materials.

(f) Prepare process sheet for the manufacturing of "Cover".



Bill of material :

Part No.	Name	Raw material	Quantity
1	Base	C.I – Casting	1
2	Thrust ball bearing	Std. Component	1
3	Spacer	C.I – Casting	1
4	(Radial) ball bearing	Std. Component	1
5	Shaft	MS φ 63	1
6	Cover	C.I – Casting	1

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